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Xanadu unveils open source machine learning software for quantum computers

Xanadu has announced the release of an open source machine learning (ML) software – PennyLane—for quantum computers.

PennyLane will allow programmers, researchers and enthusiasts globally to take part in the cutting-edge field of quantum machine learning. This software can easily integrate with currently available APIs and quantum hardware from the biggest players in the field.

Xanadu recognised the need to implement machine learning algorithms that could lead to quantum computers growing in size and therefore PennyLane was developed, envisioning it as the 'TensorFlow of quantum computing'.

PennyLane's core feature is that it implements a version of the backpropagation algorithm – the workhorse for training deep learning models – that is naturally compatible with quantum devices.

Quantum computers are getting more powerful with time, which means new quantum hardware will be available in the market and PennyLane will integrate with them easily. Moreover, PennyLane can interact with quantum computers from different companies (such as the IBM Quantum Experience) in a hardware-agnostic way because of the plugin system. It can even combine quantum sub-routines from different hardware together into a larger hybrid computation environment.

"Deep learning libraries like TensorFlow and PyTorch opened up artificial intelligence to the world by providing an interface to powerful GPU hardware. With PennyLane, Xanadu is now doing the same for machine learning on quantum hardware," said Seth Lloyd, Xanadu's chief scientific advisor, MIT professor and a founding figure in both quantum computing and quantum machine learning. "We're going to see an explosion of ideas, now that everyone can train quantum computers like they would train deep neural networks," he added.

Xanadu is a photonic quantum computing and advanced artificial intelligence company based in Toronto. It designs and integrates quantum silicon photonic chips into existing hardware to create truly full-stack quantum computing.

Drone.io announces free CI/CD cloud service for open source projects

Drone.io, the maker of the open source Drone CI/CD tool, has announced a new continuous integration/continuous delivery (CI/CD) cloud service – Drone Cloud. The company is making it available for free to open source projects. For this, it has tied up with Packet, a provider of cloud and edge infrastructure, which will run this service for free on its servers.

Brad Rydzewski, co-founder, Drone.io, said that the company's goal is to help automate the workflow of developers from the testing stage to the final release.

As part of this approach, the new Drone Cloud offers a publicly hosted offering of the CI/CD cloud. The company has teamed up with Packet to provide the required infrastructure for the new solution.

Packet provides a range of server options with varied operating systems and chips. It will make the newest Intel Xeon Scalable, ARM-based servers and AMD EPYC available to Drone Cloud service users for free as part of its multi-year donation to support the project. "Packet is offering a considerable multi-year donation to help Drone Cloud and therefore help any open source project with a GitHub repository," Rydzewski stated in a blog post.

In a statement, Jacob Smith, co-founder and CMO, Packet, said, "As open source software is deployed to increasingly diverse environments, from traditional data centres to cars and even drones, the need for multi-architecture builds has exploded."

The new CI/CD cloud service will support CD pipelines on bare metal x86, 64-bit and 32-bit ARM servers.